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REVIEW

Zebrafish, a biological model for pharmaceutical research for the management of anxiety

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Abstract
The zebrafish (*Danio rerio*) is a valuable animal model rapidly becoming more commonly used in pharmaceutical studies. Due to its low-cost maintenance and high breeding potential, the zebrafish is a suitable substitute for most adult rodents (mice and rats) in neuroscience research. It is widely used in various anxiety models. This species has been used to develop a conceptual framework for anxiety behavior studies with broad applications in the laboratory, including the study of herbal and chemical drugs. This review discusses the latest studies of anxiety-related behavior in the zebrafish model.

Keywords Zebrafish · Anxiety disorders · Behavioral tests · Neuroscience

Introduction

Traditional methods of discovering small molecular drugs were based on test results and errors of chemical compounds on phenotypical effects in cells or animals [1]. New rules and principles to reduce the use of animals for scientific experiments have been previously proposed by the European Union under the recently approved Directive 2010/63/UE [2]. Consequently, it is widely known that the number of successful new drugs on the market decreases as the expense of creating new medicines rises [3]. This problem is especially acute with drugs targeting the central nervous system (CNS) [3–5]. Anxiety is a common psychological and mental health disorder worldwide, leading to significant neurological damage [6–8]. Some anxiety disorder (AD) medications need further study to improve their healing and side-effect characteristics. Several anxiolytics, such as selective serotonin reuptake inhibitors, are also restricted by their adverse impact on sexual dysfunction and weaker therapeutic effects [8]. Medical research for nervous system disorders is expensive and risky for these reasons, such as lengthy trial periods, low success rates [9], patient heterogeneity, an absence of biomarkers, admitted patients at the initial point of disease progression [10].

Neural circuits in vertebrates have been significantly conserved during the evolution process; therefore, human fear and anxiety can be investigated by studying animals [11]. Research of rodents' anxiety-related behaviors in controlled laboratory conditions contributes to the general knowledge of normal and abnormal human anxiety [12]. Laboratory anxiety research using rodents focuses primarily on ethologically appropriate behavioral paradigms, especially for evaluating the effectiveness of medications used to treat human anxiety [13]. However, researchers using traditional behavioral techniques face two significant obstacles in rodents. First, they usually depend on anthropomorphism, which implies how circumstances impact animals' mental reactions. This issue leads to intrinsic rationality that can make replicating sophisticated analytical even among scientists who work in the same lab. Second, rodent behavioral experiments have no clear human comparisons, possibly restricting their translational interest [13]. Despite the fact that we shall address these issues in zebrafish shortly. As a result, using animals such as rats and mice to investigate

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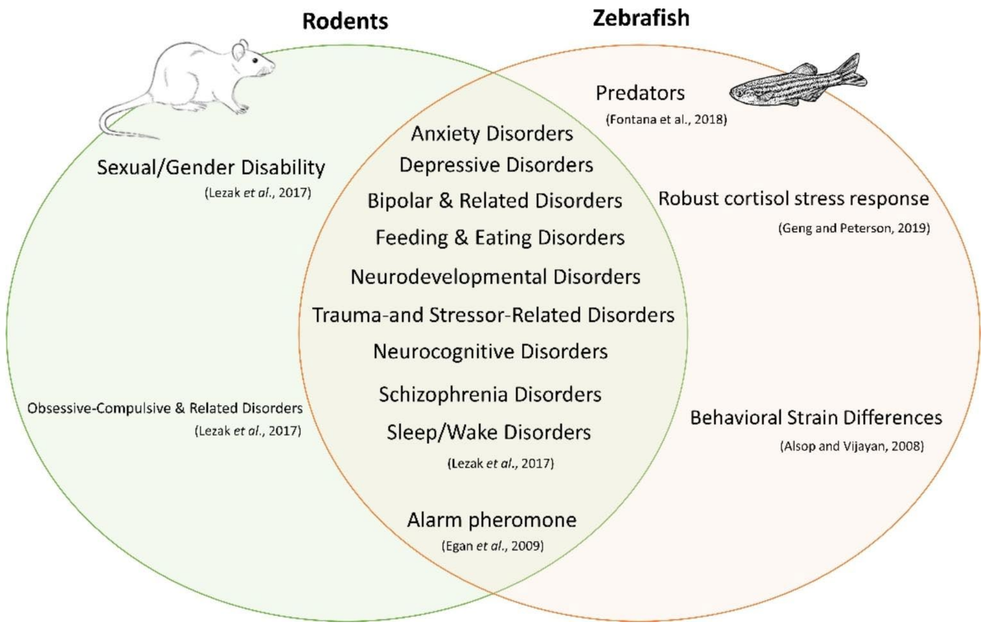

Fig. 1 Some similarities and differences in mental behavior assays in rodents and zebrafish

the causes and therapies for psychiatric disorders is difficult: the signs and symptoms of such disorders frequently represent motives, feelings, and thought processes that cannot be attributed to these organisms [13]. Also, there are a vast number of herbal medicinal products whose therapeutic ability has been tested in different animal models for measuring depression and anxiety [5].

While preliminary studies characterized fish behavior as simplistic and stereotyped, recent studies suggest complicated, context-dependent zebrafish behavioral responses [8]. Recently, interest in using zebrafish (*Danio rerio*) as a successful and high-profile anxiety model has increased [14]. This interest is possibly due partly to the assumption that fish are relatively "simple" animals, unable to confuse cognitive rules [15]. Their behavior was thought to be based on primitive instincts. Recent studies have revealed the zebrafish's complexity and relationship to fear and anxiety [16, 17], which may help model psychological and neurological illnesses such as social behavior, learning, memory, anxiety, and defensiveness [18]. Ahmad et al. (2012) reported that zebrafish larvae had the same reaction as mice and humans to drugs for stress and anxiety. Zebrafish embryos and early larvae are being grown as models in biomedical research [3]. Because of their tiny size, low cost, easy breeding, and transparency of zebrafish embryos and larvae are advantageous for high-performance chemical screening [19]. As rewarding as it might be to pin down specific frameworks on the background of homology, thus supporting the evolutionary survival of anxiety circuits, it might be more relevant to use zebrafish to uncover new aspects of the processes of fear and anxiety [11]. Subjected to triggers that cause fear or anxiety, zebrafish exhibit a variety of clear-cut measurable behaviors, including dramatically reduced exploring, increased scototaxis (dark preference), geotaxis (diving/bottom dwelling), thigmotaxis (peripheral preference), freezing (immobility) and unstable movements (rapid bouts of rapid successive turns) [8]. Moreover, zebrafish are a kind of vertebrate that is physiologically similar to humans and possesses major neurotransmitters, hormones, and receptors [8]. Thus, zebrafish studies have provided an appealing biomodel for assessing the characteristics of different agents involved in human brain disorders, such as ADs, especially neuropsychiatric disorders [8]. Figure 1 shows different neuropsychiatric disorders in rodents and zebrafish., Some behavioral characteristics in zebrafish are the same as in rodents, such as predators, robust cortisol stress responses, behavioral strain differences, and alarm pheromones [8, 20–23]. However, due to physiological differences between humans and fish, there is no way to assess communication impairments in zebrafish directly [24].

This review briefly describes an overview of anxiety disorders and the zebrafish's use as a model organism for anxiety-related behavioral research. We then provide a summary of standard anxiety assays that can be used to predict human clinical outcomes more accurately and accelerate the discovery of novel medications for treating anxiety disorders.

Anxiety disorder

Over the past two decades, anxiety disorders have been the most prevalent mental and neurological disorders [25]. The past decade has seen a remarkable and valuable development of methods to analyze anxiety-related behavior and neural circuitry [13]. Our previous papers shows that anxiety is more prevalent than depression in individuals [26, 27].

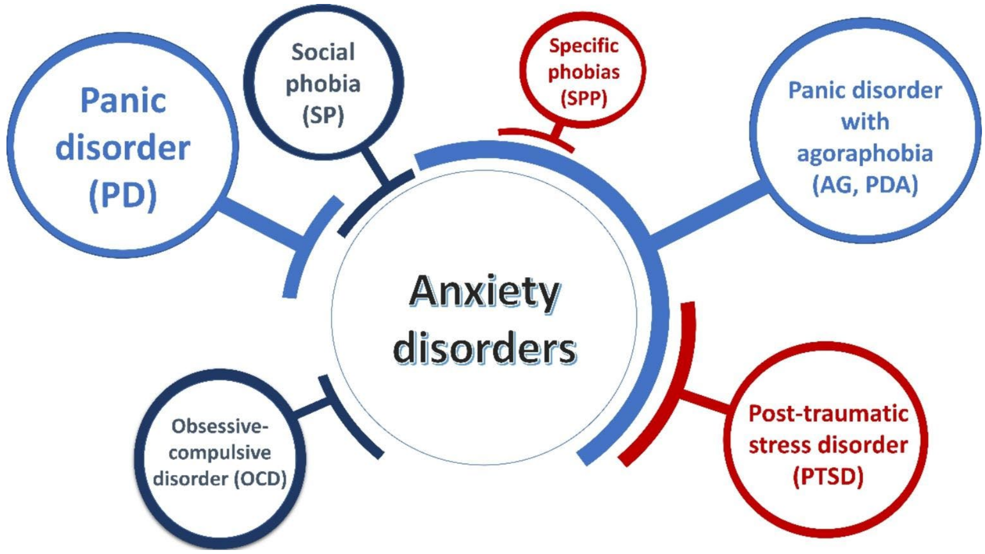

Fig. 2 Different kinds of anxiety disorders

Hence, they are often significantly related to other psychological conditions such as severe depression [28], but their pathogenicity mechanism is poorly understood [29]. Psychopharmacological and cognitive-behavioral therapies can help treat anxiety disorders [30]. Anxiety disorders affect almost 20% of adults annually [31]. Therefore, anxiety disorders burden individuals and communities [32]. Anxiety disorders may be responsible for decreased efficiency, increased morbidity and mortality rates, and increased alcohol and drug dependence in a comprehensive portion of society [30]. Anxiety disorders often coexist with other psychiatric illnesses, including drug addiction (i.e., substance use disorder) [33, 34]. It can also be pathological and may occur in problematic situations and become elevated to high levels that interfere with ordinary life [18].

Anxiety disorders encompass a diverse group of mental illnesses defined by future concerns and fear. They can lead to other psychiatric disorders, such as depression and addiction [35]. Fear and anxiety are sometimes used interchangeably, but there is a clear difference between the two. Fear is a reaction to immediate and transient risk [36] and is a more stable state resulting from risk prediction [37] or response to a possible future threat. Anxiety is frequently associated with muscle tension and avoidance behavior [38]. Besides this, critical anxiety disorders include agoraphobia, panic disorder, generalized anxiety disorder, specific phobia, social anxiety disorder associated with obsessive-compulsive disorder, post-traumatic stress disorder [39], and depression [11]. Fear and anxiety create similar cognitive reactions, such as enhanced surveillance, hypoactivity and/or freezing, increased heartbeat, and reduced meal consumption [13]. However, for the motivation of avoidance behaviors, fear and anxiety are considered critical [40]. Anxiety disorders are included in various types based on the Diagnostic and *Statistical Manual for Mental Disorders (DSM-IV)* [41], listed in Fig. 2.

The primary agents used to treat anxiety disorders are selective serotonin reuptake inhibitors and serotonin-norepinephrine reuptake inhibitors [31]. Serotonin (5-hydroxytryptamine, 5-HT) mediates inhibitory neurotransmission and excitatory and plays an essential role in cognition, anxiety, aggression, memory, training, sleep, and temperament. During growth, deterioration of 5-HT levels was associated with pathological psychiatric behaviors like disorders of the anxiety spectrum [42]. According to one of the conventional general ideas, increased 5-HT function inhibits behavioral activation, whereas decreased 5-HT function promotes behavioral activation in a variety of environmental contexts [43]. Four 5-HT receptors are found in zebrafish (*hrt1aa*, *htr1ab*, *htr1bd*, and *htr2c*) [44]. However, the neurological mechanism that underlies anxiety disorders is poorly understood; thus, new empirical approaches and theoretical concepts are needed [11]. Accordingly, there is significant interest in developing anxiety-related assays in zebrafish to identify new compounds with high potential as drugs to treat anxiety disorders.

Zebrafish as an efficient bio-model

The zebrafish is part of the Cyprinidae freshwater fish family. It includes 44 danionine species found across Southeast Asia [45]. Zebrafish are often found near rice farms, usually defined as populated water bodies that are moving slowly or standing at the margins and banks of rivers [45–47].

Modeling stress and anxiety in zebrafish has been increasingly used in neuroscience and CNS studies [8]. The introduction of efficient monitoring of zebrafish larvae and embryos has recently sparked interest in behavioral research due to its purposefulness, repeatability, and efficiency [14, 48]. Behavioral surveys of both adult fish and larvae have effectively promoted these goals. Zebrafish larvae share significant parallels with mice in terms of neuropharmacology-relevant behaviors, notably stress and anxiety, followed by adult fish and other vertebrates. [3]. For example, the larvae exhibit high cortisol concentrations in response to stress within the first weeks of development. They are susceptible to the same anxiolytics used for human anxiety management [40]. For this reason, researchers have tried to adapt many behavioral patterns of rodents, especially mice, to the zebrafish model [17], as mentioned in Fig. 1 previously.

Zebrafish larvae show a wide range of behavioral patterns, and just a few days after being laid eggs, larvae are an ideal model for behavioral and physiological screening. However, few quantitative approaches to anxiety-related responses of larvae have been reported, making it challenging to conduct coherent phenomenological analysis [49]. On the other hand, existing behavioral experiments using larvae, such as the novel tank and light-dark tests, are based on a natural aversion to light or lit areas [50]. Until recently, the behavioral endpoints in the zebrafish were handled manually, which made the data vulnerable to human error and misinterpretation. In contrast, video-tracking technology allows the behavior of animals to be analyzed automatically based on the standard observation of the endpoint of behavior, which decreases human error. The ability to store data, retrieve recorded images, and analyze video recordings is a further benefit of this data collecting technology [22].

Advantages and disadvantages of modeling anxiety in zebrafish

The zebrafish model has several unique traits that are very useful [51]. First, its transparency during the first two weeks of life allows for easy observation of abnormalities in the embryo, making these fish an ideal model for drug toxicity investigation [52]. Second, the zebrafish model is inexpensive compared to traditional in vivo models such as mice. The fish has a short life cycle, is easy to maintain, has high fertility, requires few compounds for experiments, and offers high throughput capability [53]. Therefore, a significant number of tests in a short period is feasible [54]. Third, zebrafish exhibit behavioral sensitivity to anxiety factors. The three-dimensional model with the addition of a vertical dimension can be used to detect phenotypic anxiety. In contrast, the mouse model traditionally uses two-dimensional coordinates as in a novel tank diving test that reveals common anxiety-like behaviors [55]. Also, a detailed reference guide has been developed to allow researchers to accurately identify and describe this model animal phenotype in both larvae and adults [8, 11]. Fourth, zebrafish are very similar to other vertebrates in the proper balance between system complexity and practical simplicity. Their brain anatomy, physiology, and genome are very similar to other vertebrates, including mammals [56]. We can perform this model for phenotypic screening with its broad spectrum of genetic modifications like the current genome engineering methods [57]. Thus, assessing cytotoxicity of chemical substances (chemical induction) using the zebrafish model can predict potential toxicity in humans, as the digestive, nervous, and cardiovascular systems are very similar to those of the human body [51, 58]. The high sustainability of zebrafish and mammalian therapeutic targets facilitates the laboratory use of zebrafish pharmacological research. Unlike nematodes and *Drosophila*, the genetic similarity between zebrafish and humans is at least 87% [59]. Fifth, the evaluation of zebrafish is fast. All their bodily organs mature within 36 h, and their larvae search for food and exhibit active avoidance behavior within five days after fertilization [45]. Finally, immersion makes it easy to test medicines, drugs, and compounds on zebrafish [60].

Besides these advantages, zebrafish have their limits, like many other animal models [8]. First, the solubility of test compounds is a barrier in some studies using zebrafish embryos [51], where the compounds or drugs being examined are preferably soluble in water or soluble in 1% acetone or DMSO [51, 61]. Zebrafish are tiny and cause real problems in dissection and therefore need more skill and experience in organ extraction than other typical animal models [61]. Because some zebrafish organs are fragile, it is possible to isolate them in histology [51]. Also, larvae are slightly smaller and have less pigment, making each organ hard to distinguish clearly [62]. In contrast to the mouse model, zebrafish are more expensive in manipulating gene activity [63].

Investigating the molecular mechanisms of modeling anxiety in zebrafish

One of the fundamental questions in psychiatry is how genes, molecular pathways, and binding patterns in the brain produce and modify anxiety behavior. The molecular mechanisms of anxiety and fear are strongly preserved and exist in vertebrates during the initial stages of zebrafish [40]. A wide range of mutant zebrafish strains can be used to rapidly investigate molecular mechanisms for acute neuropharmacological effects [64]. Hence, it is essential to consider complex brain disorders, including anxiety and depression, and that psychiatric disorders are essentially dependent on molecular mechanisms [42]. Lundegaard et al. (2015) discovered that treatment with MEK inhibitors acting as anticancer is effective in treating anxiety in adult zebrafish and larvae in the context of phosphodiesterase (PDE) 4 blockade/cyclic AMP (cAMP) activity [65]. Crosstalk to activation of the RAS-MAPK signal is the mechanism behind cAMP-induced anxiety [65]. Parker et al. assessed primary anxiety in zebrafish using a novel diving test. They observed that fish exposed to ethanol had reduced shoaling, social preference, and increased anxiety. This variation in behavioral activity was different in *slc6a4*, *hrt1aa*, and *oxtr* expression [66]. Furthermore, it suggests a hypothetical developmental pathway in which ethanol-induced OT inhibition leads to 5-HT suppression and up-regulation of 5-HT1A, leading to likely homeostatic up-regulation of 5-HTT at 72hpf and eventual 5-HT system imbalance [66].

Adult hippocampus neurogenesis (AHN) is also involved in multiple neurodegenerative diseases, such as Parkinson's disease, Alzheimer's disease, anxiety, and depression [67]. Campbell et al. (2004) discovered a slight reduction in hippocampal volume in human depression patients, suggesting the hippocampus is involved in mood disorders [68]. Based on some studies, the role of AHN in anxiety and depression is not entirely clear [67]. Reduced AHN in several animal models of depression confirms this discovery, as does antidepressant efficacy in restoring equilibrium. However, some polyphenols such as curcumin, catechins, and resveratrol (a compound in red grape skin) have been proven to be valuable against depression and may increase AHN, indicating that these molecules can impact behavior and not just cognition through AHN [67]. Multiple mechanisms for chlorogenic acid's anxiolytic effects have been proposed due to the variety of cellular effects revealed for this compound. Flumazenil inhibited chlorogenic acid's anti-anxiety effect, demonstrating that activating the benzodiazepine receptor reduces anxiety. Chlorogenic acid protects granulocytes against oxidative stress in a laboratory setting [69]. Also, learning and anxiety can be attributed to the particular effects of nicotine. The anxiolytic effect can be underlying by simple nicotine-induced activation of nicotine receptors. Based on zebrafish research, these varying mechanisms indicate specific treatment strategies for learning and anxiety disorders [64].

Anxiety-related behavioral tests in zebrafish

An external irritation that results in a non-specific reaction of the organism's body is called a stressor [70]. Several paradigms have been developed as practical tests to investigate anxiety-related behaviors in zebrafish that were previously related to stress-induced anxiety-related behaviors in rodents [71]. Fish are generally cultured in exposed beakers (acute therapy) or home tanks (chronic treatment) with an anxiogenic or anxiolytic drug before trying to analyze pharmacogenetic anxiety [72].

In observation tests, many fundamental behavioral models of zebrafish may be evaluated [73]. The novel tank test and the dark-light test are two well-known examples of empirical models used to study anxiety-related behaviors in zebrafish, which are evaluated based on thigmotaxis [74] and scototaxis (dark/light preference), respectively. Thigmotaxis refers to the tendency of an animal to move close to the vertical surface. The novel tank test is a comprehensive and commonly used test that identifies phenotypes of adult fish anxiety and uses the natural response of the fish to diving when situated in a new area [75]. Also, the light-dark test is considered independently as an anxiety model. However, these two tests are typically performed because no behavioral test individually can reflect anxiety, and it has been suggested that each test evaluates different aspects of anxiety [76].

In the light-dark test, a traditional behavioral assay used on rodents [77], multiple stressors may increase the time spent on the dark side; this outcome can be used to evaluate anxiety-related behaviors for drug screening [78]. Other significant objectives used in this test include the total number of transfers between the two sections, the delay in entering the white area, the frequency of risk assessment (quick entry into the light area, then revisiting the dark area or partial entry into the white area), thigmotaxis in the white area, and other endpoint behaviors such as swimming and freezing. These outcomes are similar to those recorded in the novel tank test [18, 79]. However, tests have shown that adult zebrafish avoid the light region, whereas the larvae tend to avoid the dark. Adults also avoid visiting an area near a predator [80]. They prefer to remain toward the bottom of the tank and avoid the top [80]. Additionally, zebrafish clearly show the behavioral differences in the novel tank test according to gender, age, and environmental conditions [81, 82].

Table 1 describes the types of tests related to the behavioral study of anxiety in zebrafish. The dark-light and the novel tank test behaviors exhibit mutual correlation under experimental conditions. Therefore, meta-analyses have shown that both of these tests have similar sensitivity to the modes of anxiety in zebrafish [18, 83]. However, light or darkness is not entirely neutral for some animals, but it is inherently. Studies have shown that limiting adult zebrafish to a white chamber eventually leads to a state of freezing and stagnation. However, rodents learn a constructive response to anxiety to overcome the light. In the case of anxiety in zebrafish, they are increasing the perception of the threat and affecting its priority based on the light-dark test [50].

Table 1 Tests and available models for studying anxiety-related behaviors in zebrafish

Test	Characteristics of test and its endpoints	References
Novel tank test	Based on thigmotaxis, after being acclimated to the environment, zebrafish are placed individually in a narrow tank (1.5 L) filled with water and divided into two or three equal horizontal sections. The endpoints of the test are recorded in 5–6 min. The delay in reaching the top of the tank (s), the time spent in the upper part of the tank (s), the number of times (entry) the fish entered the upper part of the tank, the number of irregular movements, the number of stagnation and freezing time (s) are recorded. In this test, anxiety is reflected in the decline of exploration, cortisol levels, thigmotaxis, and geotaxis (movement based on gravity). For example, a long delay in reaching the top of the tank, a decline in the number of entries into the upper part of the tank, a more frequent and extended time (freezing), and irregular movements and stagnation would indicate anxiety.	[84]
Light-dark test	Based on scototaxis, the endpoints of this test are delay in entering the semi-white section (s), the time spent in the semi-white section (s), and the number of entries into the semi-white section. The total time elapsed for evaluating visualization behavior can be calculated in the white section. In addition, the total number of fish rotations (90º) is divided by the total number of fish arrival times as a percentage of the rotation from the dark side of the tank.	[50, 83]
Shoaling test	Zebrafish are grouped in a ~ 40-liter tank filled with water and placed under the influence of stressor agents or conditions. Anxiogenic agents increase the coherence of fish. In contrast, by reducing anxiety, the fish exhibit a greater tendency to leave the group, and the degree of correlation of the group decreases. To perform this test, one image is recorded every 10 s. The average distance between all tested fish within the stimulated group and the mean distance traveled between the tested and the stimulated groups are evaluated. The change of medications is monitored in all tested groups (since zebrafish feel anxious, these effects are more pronounced).	[74, 85]
Social preference test	After acclimation to the new environment, zebrafish are placed in a tank full of water (e.g., 40 L) as a group. It is generally tested in huge tanks with three compartments. The model is used to evaluate the interaction mode, specifically concerning color variations. Each group can be separated from the rest by prioritizing the reflex glass, and the subject fish can be carefully observed. Adding a stressor (e.g., anxiety, stressors, or predators) causes changes in the modulus of memory and social priority.	
Open field test	After acclimation to the environment and passing the time, zebrafish are placed individually in a tank filled with water to a maximum level of 12 cm. This test is performed by recording video images used to measure all modes of discovery, thigmotaxis index, observing the relative amount of time spent around the arena wall, and endpoints. Other analyses include assessing spatiotemporal behavior and swimming in the direction of the response to the stressor.	[17, 75]
Whole-body cortisol test	This radioimmunoassay measures cortisol levels in the fish body, and it is crucial to determine the weight of each zebrafish prior to homogenization. This assay is used to evaluate the cortisol level after performing other tests such as the novel tank test and the dark-light test. This test is not considered a behavioral test, but it is included in this session because cortisol levels are based on other tests.	[86, 87]
Predator avoidance	This test evaluates the behavior associated with fear and anxiety in the presence of predators. Zebrafish is placed in the middle of a tank, and the predator is placed in one of the arms of the tank. Avoidance, or the tendency to approach the predator, is measured by the following: the delay in entering the arm where the predator is present (s), the time spent in the arm containing the predator (s), time spent in the arms without the predator (s), the number of entries into the predator's section, and the number of entries into the sections without the predator.	[88, 89]
Swimming plus-maze test	This 10-minute test is used in the early stages of growth to investigate anxiety in larvae and young zebrafish. This test is similar to the elevated plus-maze test in the rodent model. Zebrafish prefer deep arms rather than shallow and central zones during anxiety.	[49]

The shoaling test evaluates the community's general behavior in a zebrafish group [74]. Shoaling behavior is prevalent in fish models and illustrates the complicated interaction of organisms in coordinated movements —this type of fish exhibits this behavior in the wild. The open-field test is similar to the behavior paradigm in rodents. Recently, it was used in a zebrafish study focused on swimming and movement behaviors [71, 90]. In this test, cognitive factors such as habit and learning play an essential role [91].

Maze tests are another way to evaluate zebrafish behavior. Varga et al. (2018) conducted the swimming plus-maze test and found a positive and strong correlation between the selection index and the total number of arms entries. However, both variables were negative regarding the number of entries into the shallow area [49].

Utilizing physiological markers and behavioral evaluations facilitates the investigation of anxiety in zebrafish. One of these physiological factors is cortisol in the zebrafish body, as cortisol is the primary stress indicator and is typically elevated under severe or chronic stress [86]. Additionally, it may have short-term effects. Long-term increases in cortisol levels can lead to infectious diseases, defects in reproduction, and decreased growth. In zebrafish, cortisol affects the hypothalamus-pituitary-intermediate axis; in humans, it affects the hypothalamus-pituitary-adrenal axis. For this reason, physiological phenotypes significantly contribute to the use of zebrafish models in anxiety research [29]. Knowing the cortisol level in the zebrafish body provides the basis and validity of behavioral data for the researcher although it cannot be evaluated under in vivo conditions [21, 92].

However, in the latest literature related to using chemicals in anxiety-related tests in zebrafish, Biney et al. (2021) reported that 10 µM of xylopic acid decreased the duration spent at the bottom of the tank ($p < 0.001$) in the novel tank test but had no effect on the light-dark test [93]. Additionally, DePasquale and Leri found that the ability of the zebrafish to carry out bodily activities in the swimming tunnel (water flow speed of 0.5 m/s) was significantly reduced by anxiety during the 6-week experiment compared to the control group (stagnant water without activity) [94]. Also, Hamilton et al. (2017) found that a dose of 800 µM of scopolamine is an effective anti-anxiety agent for zebrafish because it triggered a significant reduction in behavior commonly associated with anxiety in zebrafish [95].

Some tips for using zebrafish in anxiety models

Despite the benefits of using zebrafish for anxiety disorder models, all animal models have limitations because they cannot thoroughly describe complex human behaviors [55]. It is difficult to differentiate between different modes of anxiety (such as generalized anxiety and fear) in different animal species. Moreover, fear in zebrafish is not fully described [96]. Another point is tracking three-dimensional motion in the novel tank test, which is further evaluated based on the measurement of the vertical displacement reaction, that is, geotaxis. While in the open field test (the primary purpose focuses on the thigmotaxis), this type of three-dimensional tracking makes it impossible [18]. However, in the open-field test, stressed zebrafish avoid swimming in the center, similar to the center-prevention of highly anxiety-sensitive humans [75].

During a group test, some fish may detach from the group. This occurs when, for example, some are more active or less anxious than others. The researcher should try to reduce environmental stress for the novel tank test. For instance, they are perhaps using similar containers for culture and testing periods and allowing the fish to recover from the transfer process (about 3 min or more) before starting the test and capturing video images [87].

According to Gerlai et al. (2009), the use of alcohol in the short-term (acute) alleviates fear and anxiety in zebrafish by increasing swimming at the top level of the tank and reducing irregular movements. It also has stimulatory effects, such as increasing the speed of movement of zebrafish and aggression [97]. There are also sex differences in the behavioral model of zebrafish, especially in shoaling behavior, but no relation to aggression. This study shows bold males possess a better shoaling tendency for unacquainted conspecifics than shy males. On the contrary, bold females are more likely than shy females to shoal and feed in the presence of stimulants. A strong relationship between shoaling and boldness, which varies according to sex, has been reported [98, 99]. For example, male zebrafish tend to be aligned with female shoals rather than males. However, the male had no preference between a mixed-gender shoal and a male- or female shoal [99]. It contradicts Gumm and colleagues' study [100], but it could help understand ecological dynamics and evolutionary repercussions for zebrafish shoaling. Overall, researchers should try to use equal numbers of fish from each sex in experiments to overcome the gender problem.

Conclusion

In summary, anxiety is a complex and multi-faceted neurobehavioral disorder, and a complete understanding can only be achieved through different approaches. Studies of anxiety-related disorders continue in the zebrafish model, with a high potential to answer any questions via analyzing neural biomarkers and the genomic index and by describing nerves based on fish-related behaviors. The behavioral patterns of zebrafish provide a stable model for parallel clinical and animal experiments, which can be used to identify pathways involved in the regulation of anxiety and to discover potential new classes of psychopharmacological drugs.

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Declarations

Competing interest The authors declare that there is no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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